Project Proposal: Contract Risk Red-Flagger

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The problem

Most small business owners sign contracts they don't fully understand. Having a lawyer review a vendor or partner agreement can cost hundreds of dollars, so people skim it and sign. The trouble shows up later: a contract renews for another year because nobody caught the 60-day cancellation window, a liability cap means they can recover almost nothing when the other side messes up, or the IP for work they paid for turns out to belong to someone else. These terms are usually in the document. They're just buried on page 14 in language nobody wants to read.

What we're building

A tool that reads a contract for you and points out what's worth worrying about. You upload a contract, and for each important clause it shows you the exact text, explains it in plain English, rates how risky it is, and suggests fairer wording you could ask for. It also tells you when something important is missing, like a contract with no cap on liability at all. It isn't a lawyer. It's a way to know which few paragraphs deserve a closer look before you sign.

You could paste a contract into ChatGPT and ask what's risky, but the answer is unstructured, can misquote clauses or invent ones that aren't there, and won't reliably notice what's absent. **Our value-add is coverage, grounding, and consistency:** every clause type is checked, every quote is verified against the real text, and risk is judged against the same written rules every time.

How it works

1. **Split and index.** Break the contract into sections and store them in a search index (sentence-transformers + FAISS).
2. **Find each clause.** For each clause type, pull the few sections most likely to contain it, then ask an LLM to quote the exact clause or say it isn't there.
3. **Check the quotes.** Confirm every quote appears word for word in the contract, and throw out anything the model made up.
4. **Explain, rate, rewrite.** For each clause, the LLM writes a plain-English summary, gives a low/medium/high rating with a reason based on our risk rubric, and drafts a fairer version.
5. **Show it.** A FastAPI backend and a simple upload page that lists the flagged clauses, color-coded by risk.

The data

We're using **CUAD** (Contract Understanding Atticus Dataset): 510 real commercial contracts filed with the SEC, where law students, reviewed by lawyers, highlighted over 13,000 clauses across 41 types (CC BY 4.0; GitHub, paper). It's our answer key: for every contract, it tells us where each clause is or that it's absent. We'll focus on a subset of clause types that carry the most risk: liability caps, uncapped liability, renewal terms, renewal notice periods, termination for convenience, non-compete, exclusivity, IP ownership, governing law, and anti-assignment. CUAD doesn't say whether a clause is good or bad, so the risk rubric and its test set are ours to build.

How we'll know it works

We'll run three systems on the same 100 held-out contracts: a model trained on CUAD's labels (the traditional ML approach), an LLM given the whole contract in one prompt (the off-the-shelf approach), and our pipeline. For finding clauses we'll measure precision and recall per clause type, how often each system correctly says a clause is missing, how often it quotes text that doesn't exist, and cost and time per contract. For the parts CUAD doesn't cover, we'll hand-label about 50 clauses to check the risk ratings, and use an LLM judge, spot-checked against our own grading, to score the explanations and rewrites.

Plan of action

1. **Data and scope.** Load CUAD, split it into train and test sets, and select the clause types the system will cover. Define a risk rubric that sets out what makes a clause high, medium or low risk.
2. **Retrieval and extraction pipeline.** Search each contract for relevant sections, extract the clauses of each type, and check that every extracted quote appears word for word in the original text.
3. **Risk analysis.** For each extracted clause, generate a plain-language explanation, assign a risk rating, and potentially suggest a safer rewrite. Flag important clauses that are missing from the contract.
4. **Evaluation.** Hand-label a test set of clauses for risk, then measure the full pipeline against the baselines on clause detection, quote accuracy and risk rating.
5. **Delivery.** Package the system as a web app, write up the methods and results, and prepare a live demo.